

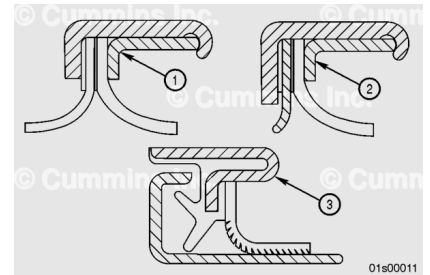
Trainings ▾

General Information

There are two types of seal used on KV engines. A one piece (1,2) and a two piece (3), unitized, seal design.

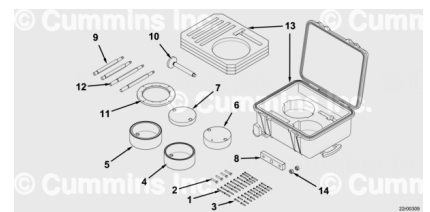
The one piece dual lip seal (1) is used in engines with wet type flywheel housings. The one piece (2) or the unitized (3) single lip seal is used in engines with dry type flywheel housings.

Note : The one piece single lip seal (2), was replaced by the unitized seal (3) in August 2010. Engines built after August 2010 will have the unitized seal installed.



The unitized rear crankshaft seal requires the rear crankshaft seal replacer kit, Part Number 5394933, for the seal removal and installation.

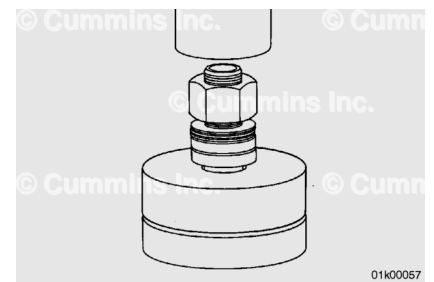
The tool consists of the following components:



1. Self tapping metal screw
2. Hexagon head capscrew
3. Self tapping metal screw
4. Crankshaft seal installer
5. Crankshaft seal installer
6. Crankshaft seal guide
7. Crankshaft seal guide
8. Installer plate
9. Mounting spacer
10. Installer screw
11. Remover plate
12. Mounting spacer
13. Storage case
14. Regular hexagon nut

Two revised service replacement rear crankshaft oil seal and wear sleeve assemblies were released for both wet and dry applications.

The revised service oil seals and thick wall (1.57 mm [0.062 in]) wear sleeves are supplied as an assembly for use on crankshafts which are grooved in the seal contact area. The revised seal and wear sleeve assembly supersedes the thin wall (0.30 mm [0.012 in]) wear sleeve, which was used with the standard service rear seal.



The revised service seal is **not** interchangeable with the standard service oil seal since it has a larger inside diameter and fits **only** the new thick wall sleeve. The standard wall sleeve has been retained for use on crankshafts that do **not** require a wear sleeve. The seal and wear sleeve **must** be installed as an assembly in one operation. If either the seal or wear sleeve requires replacement or the seal is removed from the wear sleeve during engine repair, the complete seal and wear sleeve assembly **must** be replaced. Service tool, Part Number, 3824424, is required to install the seal and wear sleeve assembly. The assembly can **not** be installed without the service tool.

The kit installs wet or dry seal assemblies to the proper depth.

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the starter air. Refer to Procedure 012-022 in Section 12.
(/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Remove the transmission, clutch, and all related components. See the equipment manufacturer instructions.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-005-tr.html)

- Remove the flywheel dowel pin from the crankshaft. Use dowel pin puller, Part Number 3163720.

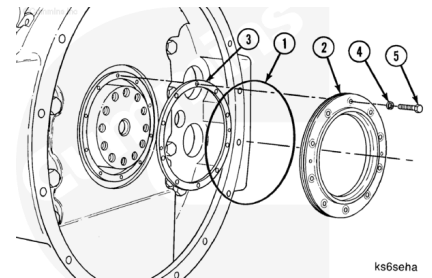
Remove

Remove the capscrews (5), washer (4), retainer (2), o-ring seal (1), and gasket (3).

Remove the crankshaft seal (**not** shown) from the retainer.

Discard the seals and the gasket.

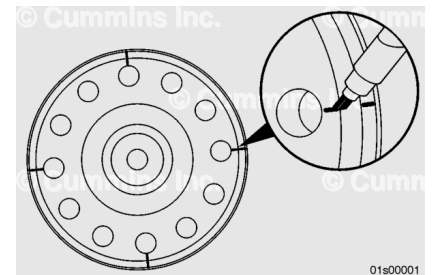
Note : The rear seal retainer can be installed in either a rear seal housing or the rear gear drive lower unit. The flywheel housing without rear seal does **not** contain the retainer.



Engines without a seal retainer use a rear crankshaft seal replacer kit, Part Number 5394933, or equivalent, to remove the seal.

Mark the seal in four positions 90 degrees apart.

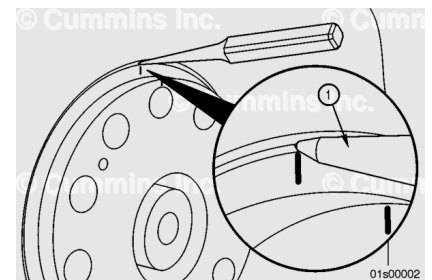
Note : Use the flywheel bolt hole locations as a guide.



Use a hammer and center punch, or similar, to distort the seal inner housing as shown in the illustration.

Note : The point of the center punch must be located between the inner and outer seal housings.

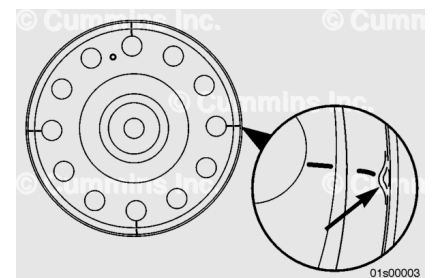
Note : Use a center punch with the shank ground down to an approximate 30 degree angle, as shown (1)



Once the step above is complete, the seal inner housing should be distorted as shown in the illustration.

This allows for access of the self tapping screws required for seal removal.

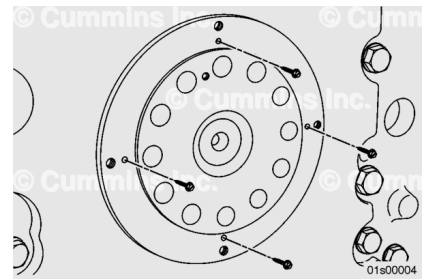
Note : If a one piece seal is installed in the engine, the seal housing will need to be drilled to allow access for the self tapping screws. Use a drill size of approximately 3 mm. Use the four 5 mm holes in the seal remover plate as a guide to indicate where to drill.



Make sure to drill straight to prevent contact with the crankshaft or flywheel housing.

Place the seal remover plate over the crankshaft with the stepped side facing toward the cylinder block.

Insert the four self tapping screws through the seal remover plate and into the seal in the four locations prepared above.

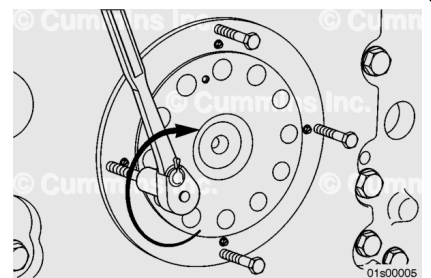


Insert the four extractor capscrews into the seal remover plate.

Turn the capscrews clockwise, alternatively and evenly, to pull the seal from the crankshaft and flywheel housing.

With the seal removed from the engine, remove the seal from the seal remover plate.

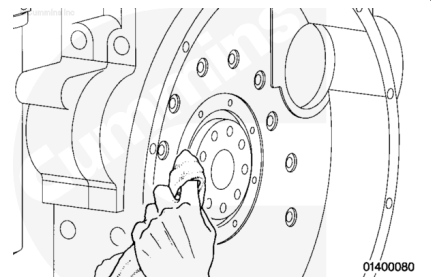
Discard the seal.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

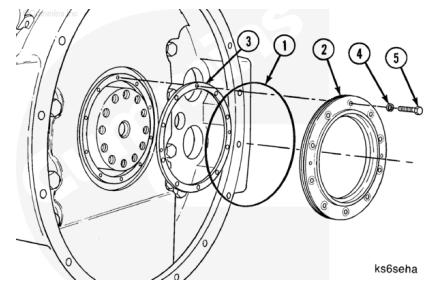


Use a clean cloth and solvent to clean the crankshaft and flywheel housing seal mounting area.

Install

Note : Not all engines have a seal retainer. Only engines that have a rear seal housing or rear gear housing have the rear seal retainer.

Note : The o-ring seal (1) **must** be installed on engines that are equipped with a wet flywheel housing. The retainer (2) for a wet housing has a groove for the o-ring. An engine with a dry flywheel housing does **not** have the o-ring seal or the retainer with a groove.

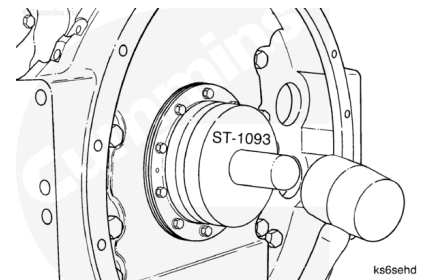


Install the o-ring (1) on the retainer (2).

Use vegetable oil to lubricate **only** the o-ring seal.

Install the gasket (3) on the retainer (2). Install, but **only** hand-tighten the ten lock washers (4) and capscrews (5).

Use seal driver, Part Number ST-1093. Install the driver on the crankshaft until the driver enters the bore of the retainer. The driver aligns the retainer with the crankshaft.



Tighten the capscrews.

Torque Value: 35 n•m [26 ft-lb]

Remove the tool.

Note : If a seal driver is **not** available, install the retainer and tighten the capscrews

Use a dial indicator to measure the crankshaft-to-retainer bore alignment. The retainer **must** be concentric with the crankshaft within 0.13 mm [0.005 in] total indicator run out.

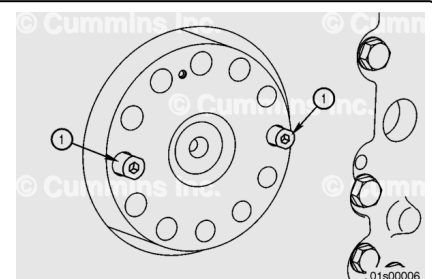
If the retainer is **not** in alignment, loosen the capscrews and move the retainer until it is within specification, then tighten the capscrews.

Torque Value: 35 n•m [26 ft-lb]

For unitized seal (dry flywheel housings), use Rear Crankshaft Seal Replacer Kit, Part Number 5394933, to install the seal.

Install the mounting adaptors (1) in to the crankshaft 180 degrees apart.

Tighten the adaptors



Torque Value:

Adapters 25 n•m [221 in-lb]

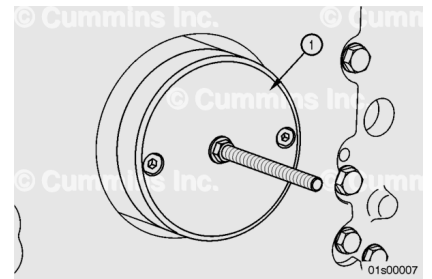
Install the crankshaft pilot (1) onto the crankshaft over the adaptors.

Install the mounting capscrews through the pilot into the adaptors.

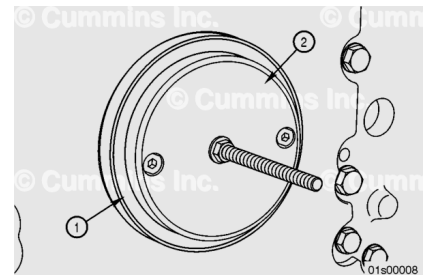
Tighten the mounting capscrews.

Torque Value:

Mounting Capscrews 15 n·m [133 in-lb]

**⚠ CAUTION ⚠**

Do not lubricate. The seal, crankshaft, and flywheel housing must be clean and dry or a leak can result.



Install the seal (1) over the pilot (2) until it contacts the crankshaft.

Note : The seal part number is printed on the front face of the seal. The part number must face away from the cylinder block once installed.

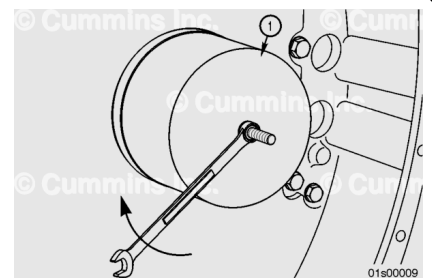
Note : Seal driver, Part Number ST-1093, must not be used to install the unitized seal.

Note : Do not use air tools to drive the seal into place.

Install the crankshaft seal installer (1) over the pilot.

Install the washer and nut onto the threaded bar.

Tighten the nut until the seal installer contacts the flywheel housing. Make sure the seal installer has full contact with the flywheel housing.

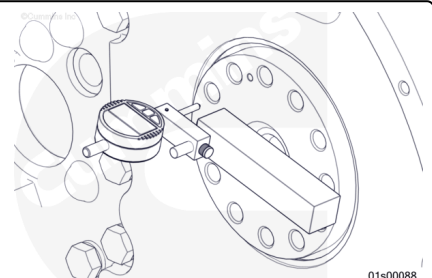
**Torque Value:**

Approximate Torque Value 60 n·m [44 ft-lb]

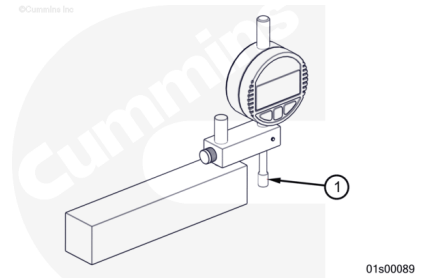
Remove the seal installer and pilot from the engine.

To check the seal has been installed correctly, Total Internal Runout measurements need to be taken using the digital depth gauge, Part Number 3377902. This will check if the seal has been installed within specification.

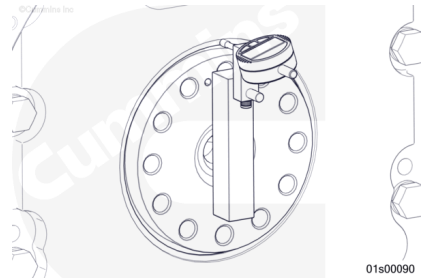
Hold the depth gauge base flat on the end of the crank shaft so that the depth gauge contact point is perpendicular to the surface of the crank shaft. Adjust the height of the gauge so that it is in the middle of its measurable range and zero the gauge.



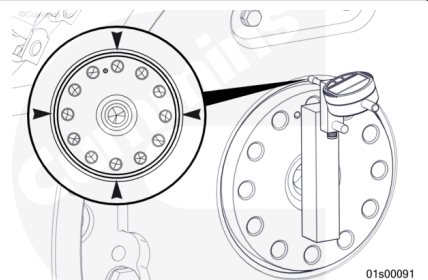
Attach the 12.7 mm [$\frac{1}{2}$ in] gauge contact point (1).



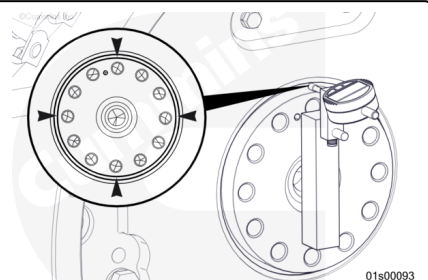
Place the depth gauge base back on the end of the crankshaft at the 12 o'clock position with the contact point applied on the stationary part of the outer seal.



Measure the installation depth of the stationary outer seal at the 12, 3, 6, and 9 o'clock positions and record it in the table below.



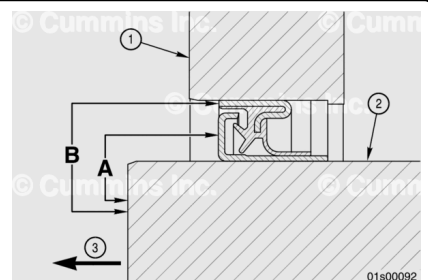
Measure the installation depth of the rotating inner wear sleeve at the 12, 3, 6, and 9 o'clock positions and record it in the table below.



Determine the Maximum and Minimum installation depths for the stationary and rotating parts of the seal and fill in the corresponding parts of the table.

1. Flywheel housing
2. Crankshaft
3. Rear of engine.

- A. Sleeve/Seal
- B. Stationary part.



Fill in the second table by completing the calculations below.

Square Stationary = Maximum Stationary - Minimum Stationary

Square Rotating = Maximum Rotating - Minimum Rotating

Step Maximum = Rotating Minimum - Stationary Maximum

Step Minimum = Rotating Maximum - Stationary Minimum

Installation Depth	12 o'clock position	3 o'clock position	6 o'clock position	9 o'clock position	Max	Min	Installation Tolerance
Stationary							1.05±0.5 mm [0.413±0.0196 in]
Rotating							1.05±0.5 mm [0.413±0.0196 in]

Note : The tolerances contained in this table are to be used with the digital depth gauge with 12.7 mm [½ in] extension.

Measurement Location	Acceptable Tolerance	Calculated
Square Stationary	< 0.5 mm [0.0196 in]	
Square Rotating	< 0.5 mm [0.0196 in]	
Step Maximum	< 0.5 mm [0.0196 in]	
Step Minimum	< 0.5 mm [0.0196 in]	

If the seal is within specification, continue to the finishing steps.

If the seal is out of specification, it may malfunction after a low number of hours in service. The seal will have to be removed and a new seal installed.

Note : If the seal is installed out of specification, the tool may be worn and require replacing.

If the seal is within specification, continue to the finishing steps. If the seal is out of specification, it may fail after a low number of hours in service. The seal will have to be removed and a new seal installed.

Note : If the seal is installed out of specification, the installation tool may be worn and require replacing.

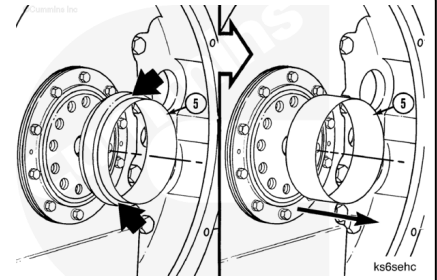
Seal	Installation Depth	Tolerance
Stationary Seal	13.75 mm [0.541 in]	±0.5 mm [±0.0196 in]
Rotation Seal	13.75 mm [0.541 in]	±0.5 mm [±0.0196 in]

For the dual lip seal (wet flywheel housings), use the installation sleeve (5) that is with the seal.

Push the seal from the sleeve and onto the crankshaft. Push the seal on the crankshaft until it touches the retainer or the housing.

Remove the installation sleeve (5).

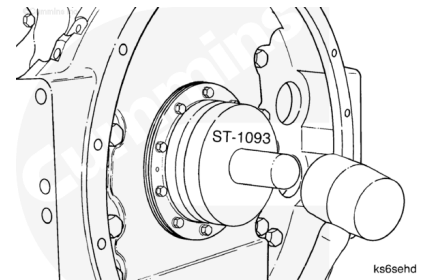
Note : The dual lip seal **must** be installed with the black element toward the cylinder block.



Use the seal driver, Part Number ST-1093, or equivalent, and a mallet. Install the seal in the bore until the driver spacers contact the crankshaft.

Remove the tool.

Note : Rear crankshaft seal replacer kit, Part Number 5394933, can be used to install the dual lip seal as described in the unitized seal steps above.

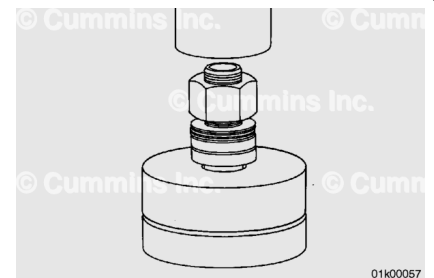


Note : The revised service seal is **not** interchangeable with the standard service oil seal since it has a larger inside diameter and fits **only** the new thick wall sleeve. The standard wall sleeve has been retained for use on crankshafts that **do not** require a wear sleeve.

The seal and wear sleeve **must** be installed as an assembly in one operation. If either the seal or wear sleeve requires replacement or the seal is removed from the wear sleeve during engine repair, the complete seal and wear sleeve assembly **must** be replaced.

Service tool, Part Number 3824424, is required to install the seal and wear sleeve assembly. The assembly can **not** be installed without the service tool.

The kit installs wet or dry seal assemblies to the proper depth.



Finishing Steps



This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

** WARNING **

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the flywheel dowel pin and flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/28/28-016-005-tr.html>)
- Install the transmission, clutch, and all related components. See the equipment manufacturer instructions.
- Connect the starter air. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Connect the batteries. See equipment manufacturer service information.
- Start the engine. Check equipment operation.